

Critically discuss the Allais Paradox

The Allais Paradox starts by considering the choice between two lotteries A and B. For lottery A, there is a guaranteed chance of winning 1 million dollars, and for lottery B there is an 89% chance of winning 1 million dollars, a 10% chance of winning 5 million dollars, and a 1% chance of winning nothing.

We then consider the choice between lotteries C and D. For lottery C, there is an 11% chance of winning 1 million dollars, and an 89% chance of winning nothing. For lottery D, there is a 10% chance of winning 5 million dollars and a 90% chance of winning nothing.

I think that most people would choose A between the over B because, despite the greatest monetary value being much greater for B than A, the subjective utility for, for most people, is not so different. This is because each successive dollar gained by an actor decreases in subjective value - this is the concept of marginal utility in economics. Given that the greatest winnings of A and B are now a little more comparable, it would be irrational to expose yourself to the risk of not winning anything at all.

We could, however, conceive of a case where the difference in the monetary value is so much greater, such as 9 million dollars instead of 4 million dollars, such that the subjective utility differs more greatly and 1 million dollars and 10 million dollars become very different in our minds.

In the case of lotteries C and D, the difference between the lotteries of winning nothing is only 1%, but the difference between the greatest monetary values remains the same. I think that most people would probably find the 1% easy to dismiss due to the lack of certainty where it usually over-weighted. Therefore, they choose lottery D despite being risk averse in choosing A over B.

The choices of both A and D are called the Allais preferences. These preferences are at odds with strong independence because there is an identical 89% for A and B, and C and D which should not affect the decision.

Emotions

Since utility is a subjective quantity which is supposed to explain the behaviour of an actor, and an actor perceives risks and gains through the lens of their psychology, I think that utility should consider the emotions of the actor. Therefore, every outcome in lotteries B, C, and D which is a zero-dollar winning is replaced with a term denoting the negative utility value of the negative emotion. Weber (1998, pp. 102-105) evaluates two candidate emotions which may justify the Allais preferences.

The first is *disappointment*, which the actor would feel if they chose B over A, only to gain nothing. The slimmer the chance of winning nothing and the greater the potential reward, the greater the disappointment. The disappointment of not having won anything upon choosing D would be far less than for B because the actor understands it is more likely that they may not win anything. According to Weber, this solution fails because, although the disappointment associated with lottery B is greater than that of lottery D, lottery D's disappointment is more likely – and we require that the lottery B is more disappointing to justify the Allais preferences. We would have to conclude that rational actors must judge for themselves as best they can whether the disappointment whether the disappoint from lottery B would be ninety times as intense as lottery D.

I disagree. It seems entirely possible to me that a rational actor could judge for themselves, in some cases, whether one disappointment is more intense than another – especially if they have had similar experiences. Although perhaps a less 'nice' number such as eighty-seven instead of ninety for a modified Allais Paradox would be even more difficult for a rational actor. I doubt that this increased difficulty in consideration would be reflected in the increased time taken for rational actors to respond to the paradox.

The second emotion Weber considers is *regret*. The greater the monetary outcome would have been if the rational actor had chosen differently, the greater the disutility of regret. Unlike disappointment, regret is complementary, meaning that our emotions do not evaluate our options independently of each other. Regret is greater for B than it is for D because there is a guaranteed 1 million dollars to regret if the rational actor wins nothing in choosing B, while C still offers an 89% chance of winning nothing.

Limitations of the Emotional Approach

I would not require that a rational actor precisely state the utility of the winnings and the disappointment in the same units. But I think that the use of emotions makes the problem even more difficult for the rational actor than Weber has let on. Introducing emotions into the payoffs would contaminate objective, determined, quantities with subjective and unreliable ones, causing rational

actors to be unable to understand their own subjective utility function.

This is because emotions, unlike monetary value, must be estimated before making the gamble. These estimates are based on, not the monetary value itself, but rather the values of subjective utility function of the winnings. Unfortunately for the rational actor, it becomes increasingly difficult to understand even roughly how disappointed they should be or would be as the winnings of the Allais paradox are increasingly modified – therefore affecting the subjective utility of the winnings.¹

It would be dishonest to introduce negative emotions to lottery payoffs to resolve the Allais Paradox without introducing positive emotions. What if we include an emotion which makes the lottery with the greatest monetary value more attractive? Actors should be rational, but if they can be biased by the prospect of disappointment or regret, I fail to see why they would be beyond the rush of maximising their payoff. 'Rush' is an emotion with positive utility which is added to both 5 million dollars offered by B and D. Rush is greater for D because it is riskier, with only a 10% chance of winning any money. The rush of choosing B may not be sufficiently enticing as the sensible choice which guarantees 1 million dollars, while lottery D is still attractive.

This emotion explains the Allais preferences, but at the cost of making the Allais preferences seem more irrational.

Conclusion

I think that the problem with using emotions to resolve the Allais Paradox is that it becomes difficult to establish when an emotion becomes irrational. Surely rush is irrational, it is the basis of gambling – which no rational actor must ever engage in. However, perhaps it is not irrational for the gambler if their honest subjective utility function assigns more utility to the rush itself rather than the winnings, or at least, the expected losses. Entirely rational actors may therefore be likely to gamble socially or drink and smoke socially in an attempt to offset the disutility.

REFERENCES

Weber, M., 1998. The resilience of the Allais paradox. *Ethics* 109: 94–118.

1. I imagine that a diligent rational actor would have a very difficult time getting a rough idea of their subjective utility function if they were reminded before they gambled exactly how much their rent cost compared to the winnings.